Experiment 6

AIM: Create Product Catalog Application using ASP.NET MVC Architecture (Controllers, Models, Views, Routing)

Code:

Models/Product.cs

```
using System.ComponentModel.DataAnnotations;

namespace ProductCatalog.Models
{
    public class Product
    {
        [Key]
        public int ProductId { get; set; }

        [Required]
        public string ProductName { get; set; }

        [Required]
        public string Category { get; set; }

        [Required]
        public decimal Price { get; set; }

        public string Description { get; set; }

        public string ImageUrl { get; set; }
    }
}
```

Controllers/ProductController.cs

```
using System.Collections.Generic;

using System.Linq;

using System.Web.Mvc;

using ProductCatalog.Models;

namespace ProductCatalog.Controllers

{
```

```
public class ProductController : Controller
{
    // Temporary product list
    private static List<Product> products = new List<Product>
    {
        new Product
        {
            ProductId = 1,
            ProductName = "Laptop",
            Category = "Electronics",
            Price = 55000,
            Description = "High-performance laptop",
            ImageUrl = "https://via.placeholder.com/150"
        },
        new Product
        {
            ProductId = 2,
            ProductName = "Mobile Phone",
            Category = "Electronics",
            Price = 25000,
            Description = "Latest smartphone",
            ImageUrl = "https://via.placeholder.com/150"
        },

        new Product
        {
```

```
            ProductId = 3,
            ProductName = "Headphones",
            Category = "Accessories",
            Price = 2000,
            Description = "Wireless headphones",
            ImageUrl = "https://via.placeholder.com/150"
        }
    };

    // GET: Product
    public ActionResult Index()
    {
        return View(products);
    }

     // GET: Product/Details/1
    public ActionResult Details(int id)
    {
        Product product = products.FirstOrDefault(p => p.ProductId == id);

        if (product == null)
            return HttpNotFound();

        return View(product);
    }
```

```
        // GET: Product/Create
        public ActionResult Create()
        {
            return View();
        }

        // POST: Product/Create
        [HttpPost]
        public ActionResult Create(Product product)
        {
            if (ModelState.IsValid)
            {
                product.ProductId = products.Count == 0
                    ? 1
                    : products.Max(p => p.ProductId) + 1;

                products.Add(product);

                return RedirectToAction("Index");
            }

            return View(product);
        }
    }
}
```

Calendar.aspx

```
<%@ Page Title="Academic Calendar" Language="C#"
    MasterPageFile="~/Site.Master"
    AutoEventWireup="true"
    CodeBehind="Calendar.aspx.cs"
    Inherits="AcademicCalendarLeaveManagement.Calendar"  %>

<asp:Content ID="Content1"
    ContentPlaceHolderID="MainContent"
    runat="server">

    <div class="container">

        <h1>Academic Calendar</h1>

        <asp:Label ID="lblWelcome"
            runat="server"
            CssClass="welcome">
        </asp:Label>

        <hr />

        <h2>Academic Events</h2>
```

```
<!-- Rich Control: Calendar -->

<asp:Calendar ID="AcademicCalendar"
    runat="server"
    Width="100%"
    Height="350px"
    BackColor="White"
    BorderColor="#999999"
    DayNameFormat="Shortest"
    Font-Names="Verdana"
    Font-Size="10pt"
    ForeColor="Black"
    OnSelectionChanged="AcademicCalendar_SelectionChanged">

    <SelectedDayStyle
        BackColor="#666666"
        Font-Bold="True"
        ForeColor="White" />

    <TodayDayStyle
        BackColor="#CCCCCC"
        ForeColor="Black" />

    <OtherMonthDayStyle
        ForeColor="#808080" />
```

```
    <TitleStyle
        BackColor="#5A5A5A"
        Font-Bold="True"
        ForeColor="White" />

</asp:Calendar>

<br />

<asp:Label ID="lblSelectedDate"
    runat="server"
    CssClass="dateMessage">
</asp:Label>

<hr />

<!-- Rich Control: TreeView -->

<h2>Academic Information</h2>

<asp:TreeView ID="AcademicTree"
    runat="server"
    ExpandDepth="2">

    <Nodes>
```

```
<asp:TreeNode Text="Academic Year 2026-27">

  <asp:TreeNode Text="Semester 1">

    <asp:TreeNode Text="Semester Start: July 2026" />

    <asp:TreeNode Text="Mid Semester Exam: September 2026" />

    <asp:TreeNode Text="End Semester Exam: November 2026" />

  </asp:TreeNode>

  <asp:TreeNode Text="Semester 2">

    <asp:TreeNode Text="Semester Start: January 2027" />

    <asp:TreeNode Text="Mid Semester Exam: March 2027" />

    <asp:TreeNode Text="End Semester Exam: May 2027" />

  </asp:TreeNode>

  <asp:TreeNode Text="Holidays">

    <asp:TreeNode Text="Independence Day" />
```

```
                <asp:TreeNode Text="Republic Day" />

                <asp:TreeNode Text="Diwali Vacation" />

            </asp:TreeNode>

        </asp:TreeNode>

    </Nodes>

</asp:TreeView>

<br />

<asp:HyperLink ID="lnkLeave"
    runat="server"
    NavigateUrl="~/Leave.aspx"
    CssClass="btn">

    Apply for Leave

</asp:HyperLink>

</div>

</asp:Content>
```

Views/Product/Index.cshtml

```
@model IEnumerable<ProductCatalog.Models.Product>

@{
    ViewBag.Title = "Product Catalog";
}

<h2 class="text-center">Product Catalog</h2>

<p>
    @Html.ActionLink(
        "Add New Product",
        "Create",
        "Product",
        null,
        new { @class = "btn btn-primary" }
    )
</p>

<div class="row">

@foreach (var product in Model)
{
    <div class="col-md-4">
```

```
<div class="panel panel-default">

  <div class="panel-heading">
    <h3>@product.ProductName</h3>
  </div>

  <div class="panel-body">

    <img src="@product.ImageUrl"
         class="img-responsive"
         alt="@product.ProductName" />

    <p>
      <strong>Category:</strong>
      @product.Category
    </p>

    <p>
      <strong>Price:</strong>
      ₹@product.Price
    </p>

    <p>
      @product.Description
    </p>
```

```
                @Html.ActionLink(
                    "View Details",
                    "Details",
                    "Product",
                    new { id = product.ProductId },
                    new { @class = "btn btn-success" }
                )


            </div>
        </div>
    </div>
}

</div>
```

Views/Product/Details.cshtml

```
@model ProductCatalog.Models.Product
@{
    ViewBag.Title = "Product Details";
}

<h2>Product Details</h2>
```

```
<div class="panel panel-primary">

    <div class="panel-heading">
        @Model.ProductName
    </div>

    <div class="panel-body">

        <img src="@Model.ImageUrl"
             class="img-responsive"
             alt="@Model.ProductName" />

        <br />

        <p>
            <strong>Product ID:</strong>
            @Model.ProductId
        </p>

        <p>
            <strong>Product Name:</strong>
```

```
    @Model.ProductName
</p>

<p>
    <strong>Category:</strong>
    @Model.Category
</p>

<p>
    <strong>Price:</strong>
    ₹@Model.Price
</p>

<p>
    <strong>Description:</strong>
    @Model.Description
</p>

@Html.ActionLink(
    "Back to Products",
    "Index",
```

```
            "Product",
            null,
            new { @class = "btn btn-default" }
        )

    </div>

</div>
```

Views/Product/Create.cshtml

```
@model ProductCatalog.Models.Product

@{
    ViewBag.Title = "Add Product";
}

<h2>Add New Product</h2>

<hr />

@using (Html.BeginForm("Create", "Product", FormMethod.Post))
{
    @Html.AntiForgeryToken()
```

```
<div class="form-group">

    @Html.LabelFor(model => model.ProductName)

    @Html.TextBoxFor(
        model => model.ProductName,
        new
        {
            @class = "form-control",
            placeholder = "Enter product name"
        }
    )

    @Html.ValidationMessageFor(
        model => model.ProductName,
        "",
        new { @class = "text-danger" }
    )

</div>

<br />

<div class="form-group">

    @Html.LabelFor(model => model.Category)
```

```
    @Html.TextBoxFor(
        model => model.Category,
        new
        {
            @class = "form-control",
            placeholder = "Enter category"
        }
    )

    @Html.ValidationMessageFor(
        model => model.Category,
        "",
        new { @class = "text-danger" }
    )

</div>

<br />

<div class="form-group">

    @Html.LabelFor(model => model.Price)

    @Html.TextBoxFor(
        model => model.Price,
```

```
            new
            {
                @class = "form-control",
                type = "number",
                step = "0.01",
                placeholder = "Enter price"
            }
        )

        @Html.ValidationMessageFor(
            model => model.Price,
            "",
            new { @class = "text-danger" }
        )

    </div>

    <br />

    <div class="form-group">

        @Html.LabelFor(model => model.Description)

        @Html.TextAreaFor(
            model => model.Description,
            new
```

```
        {
            @class = "form-control",
            rows = "4",
            placeholder = "Enter product description"
        }
    )

</div>

<br />

<div class="form-group">

    @Html.LabelFor(model => model.ImageUrl)

    @Html.TextBoxFor(
        model => model.ImageUrl,
        new
        {
            @class = "form-control",
            placeholder = "Enter image URL"
        }
    )

</div>
```

```
    <br />

    <button type="submit" class="btn btn-primary">
        Add Product
    </button>

    @Html.ActionLink(
        "Cancel",
        "Index",
        "Product",
        null,
        new { @class = "btn btn-default" }
    )
}
```

App_Start/RouteConfig.cs

```
using System.Web.Mvc;
using System.Web.Routing;

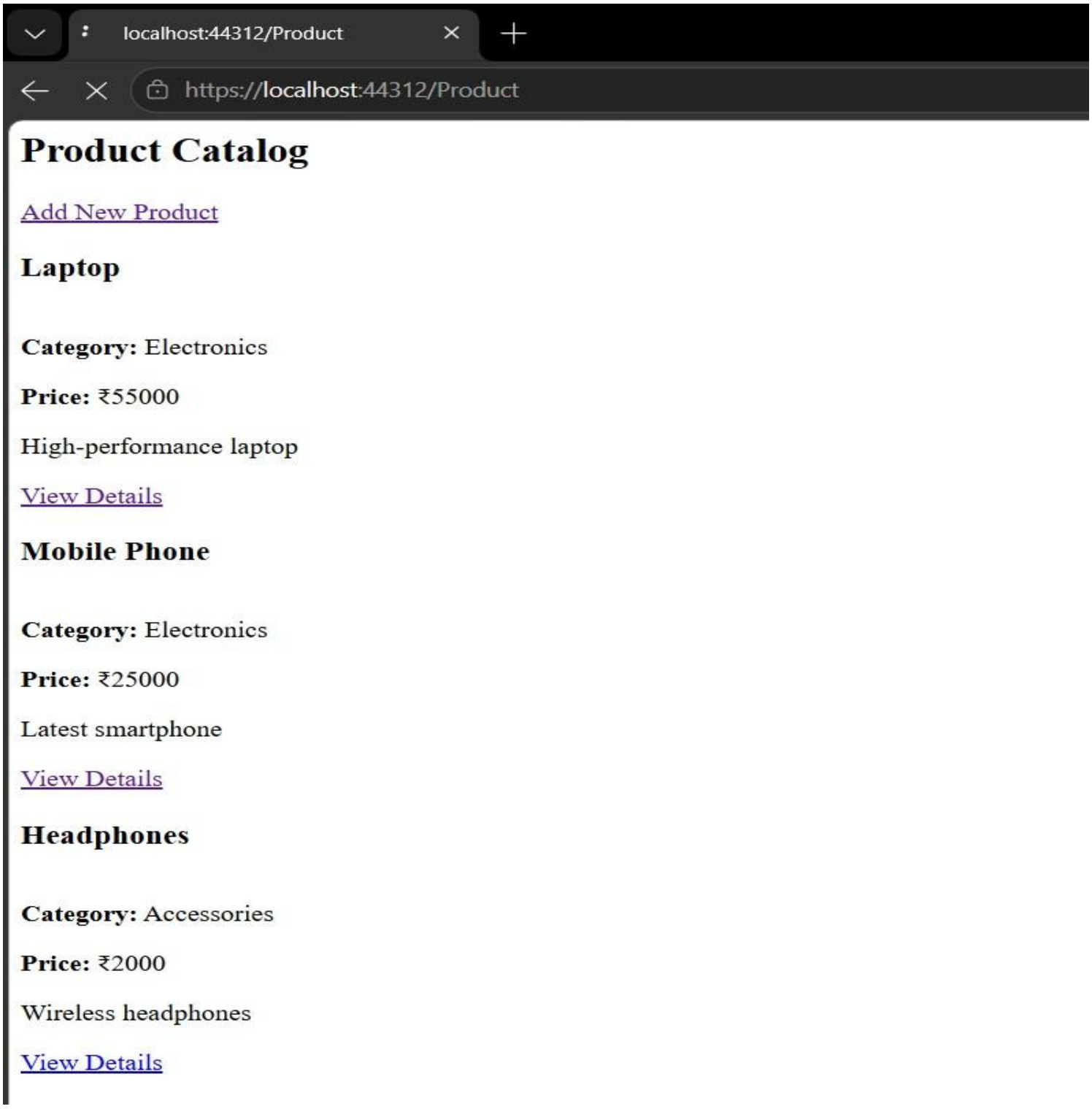
namespace ProductCatalog
{
    public class RouteConfig
    {
        public static void RegisterRoutes(RouteCollection routes)
```

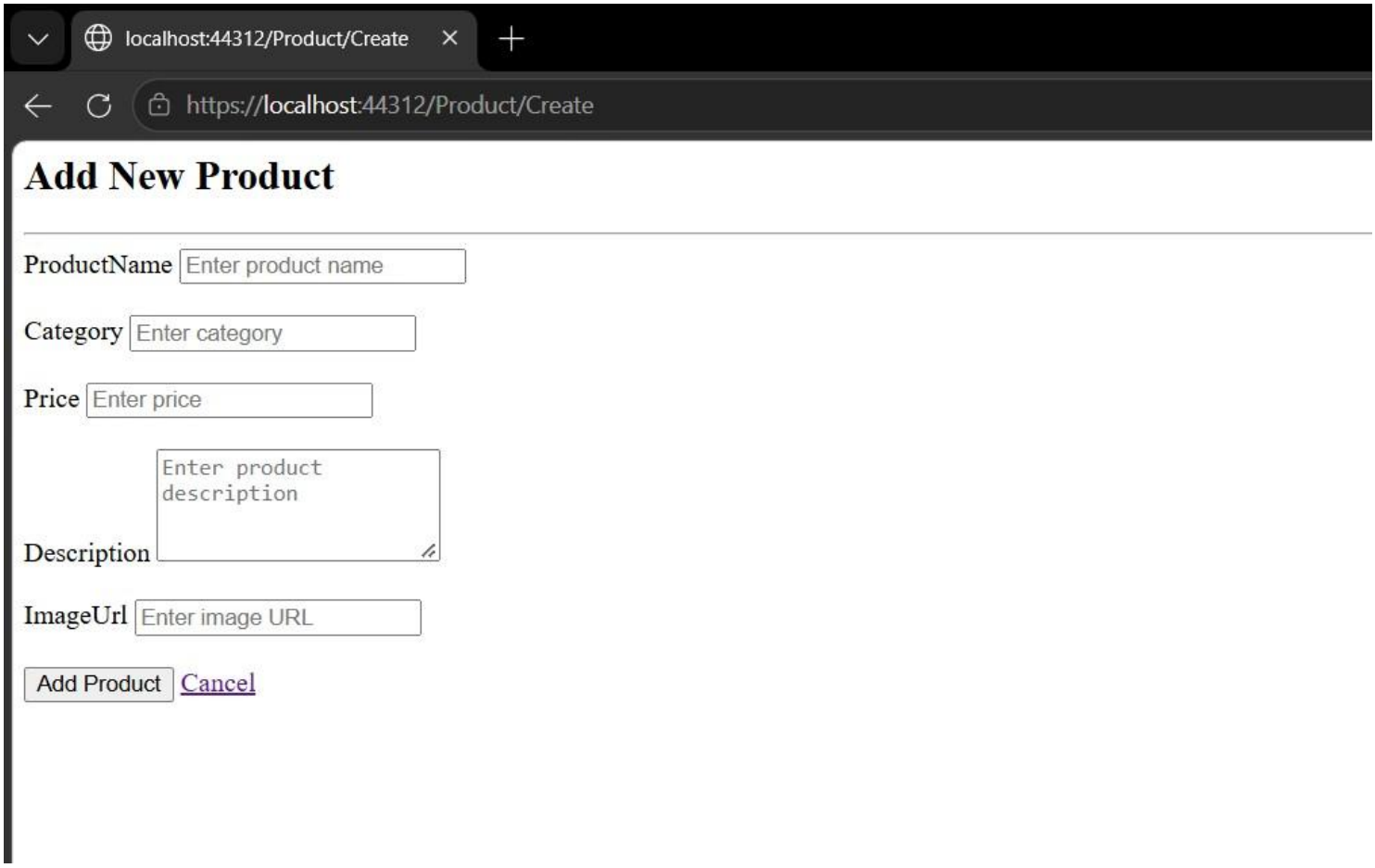
```
        {
            routes.IgnoreRoute("{resource}.axd/{*pathInfo}");

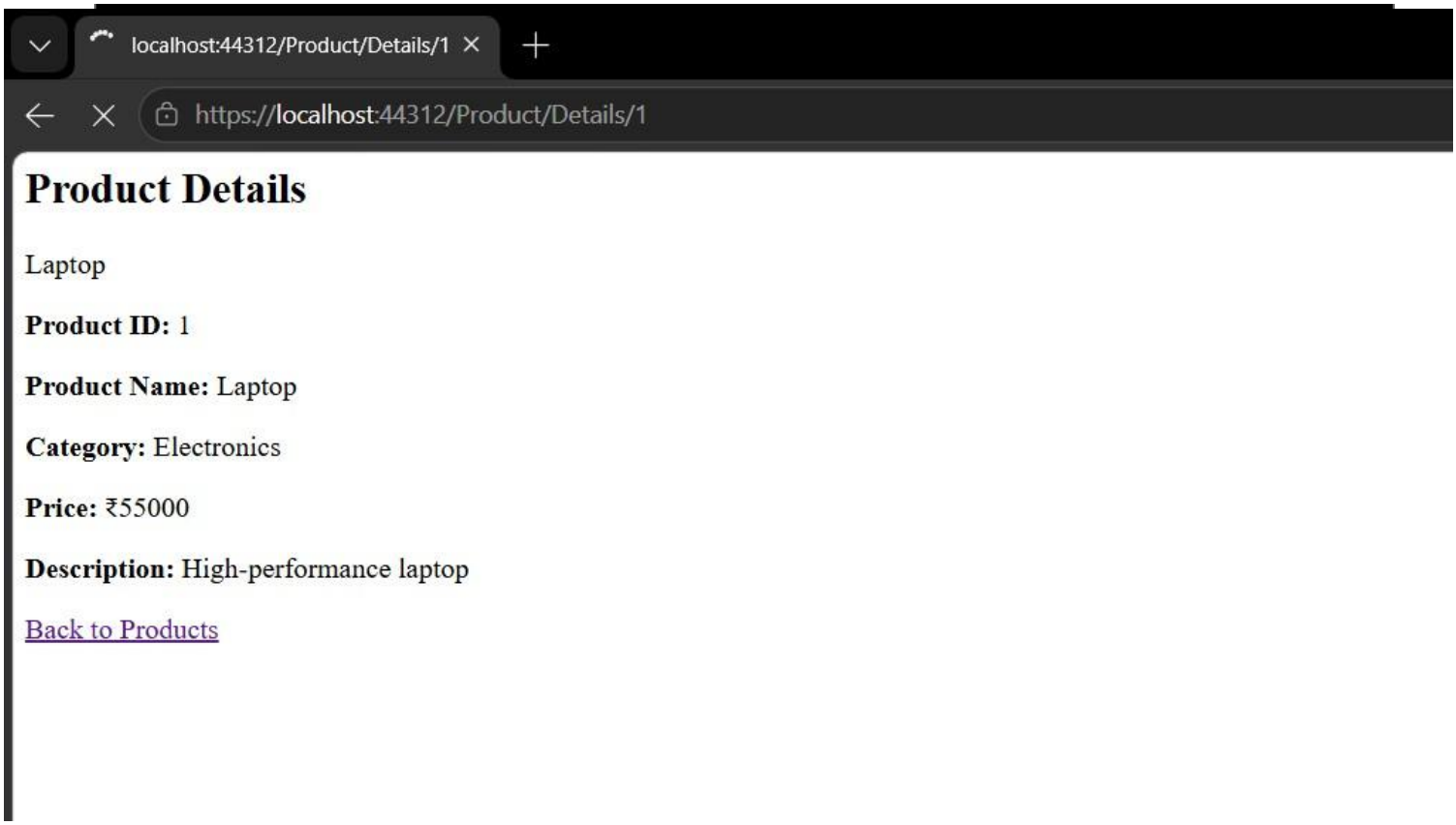
            routes.MapMvcAttributeRoutes();

            routes.MapRoute(
               name: "Default",
               url: "{controller}/{action}/{id}",
               defaults: new
               {
                  controller = "Product",
                  action = "Index",
                  id = UrlParameter.Optional
               }
            );
        }
    }
}
```

Output: Screenshot:




localhost:44312/Product/Create
https://localhost:44312/Product/Create
Add New Product
ProductName
Enter product name
Category
Enter category
Price
Enter price
Description
Enter product description
ImageUrl
Enter image URL
Add Product
Cancel


localhost:44312/Product/Details/1
https://localhost:44312/Product/Details/1
Product Details
Laptop
Product ID: 1
Product Name: Laptop
Category: Electronics
Price: ₹55000
Description: High-performance laptop
Back to Products

GitHub URL: